

QUESTION 1

Code:

```
# Load required package
library(ggplot2)

# Use the built-in Nile dataset
nile_ts <- ts(Nile, start = 1871, frequency = 1)

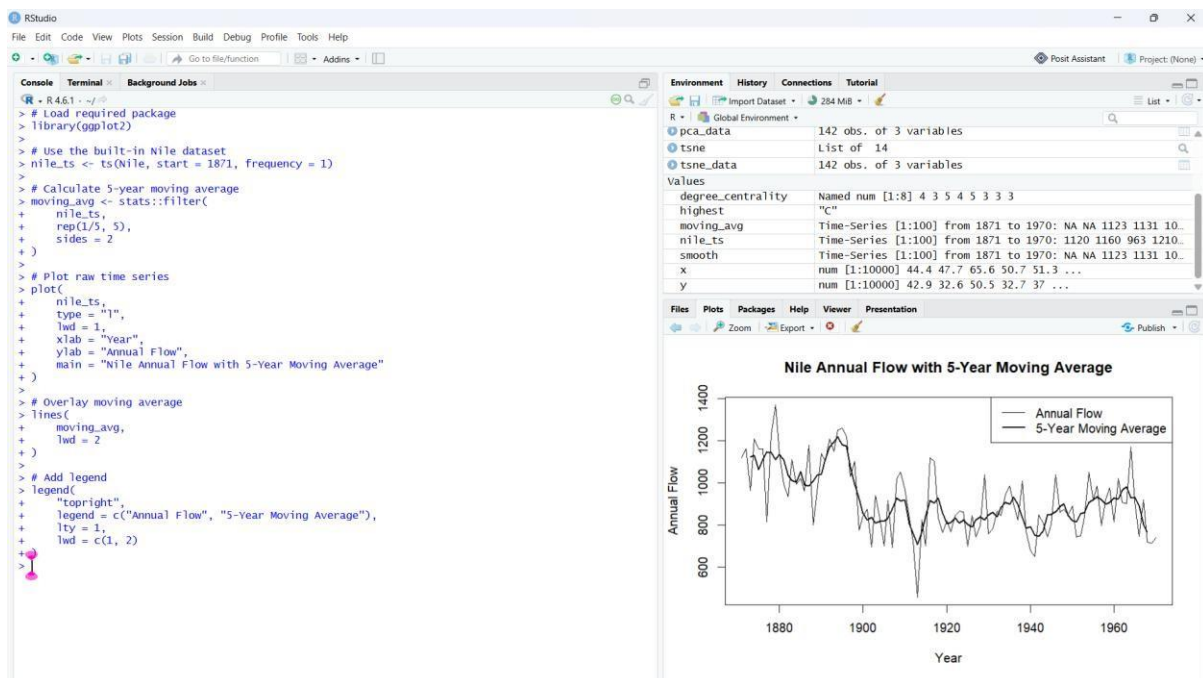
# Calculate 5-year moving average
moving_avg <- stats::filter(
  nile_ts,
  rep(1/5, 5),
  sides = 2
)
# Plot raw time series
plot(
  nile_ts,
  type = "l",
  lwd = 1,
  xlab = "Year",
  ylab = "Annual Flow",
  main = "Nile Annual Flow with 5-Year Moving Average"
)
# Overlay moving average
lines(
  moving_avg,
  lwd = 2
```

```

)
# Add legend
legend(
  "topright",
  legend = c("Annual Flow", "5-Year Moving Average"),
  lty = 1,
  lwd = c(1, 2)
)

```

Output:



Interpretation

- The raw line shows the **annual Nile river flow** from 1871 onwards.
- The 5-year moving average smooths short-term fluctuations and highlights the **overall pattern**.
- The series shows considerable **year-to-year variation** in river flow.

QUESTION 2

Code:

```
library(ggplot2)
```

```
library(tidyr)
```

```
# PART A: Multiple Time Series
```

```
stock_data <- as.data.frame(EuStockMarkets)
```

```
stock_data$time <- time(EuStockMarkets)
```

```
stock_long <- pivot_longer(
```

```
  stock_data,
```

```
  cols = c(DAX, SMI, CAC, FTSE),
```

```
  names_to = "Index",
```

```
  values_to = "Value"
```

```
)
```

```
stock_long$Index_Value <- ave(
```

```
  stock_long$Value,
```

```
  stock_long$Index,
```

```
  FUN = function(x) (x / x[1]) * 100
```

```
)
```

```
ggplot(stock_long, aes(time, Index_Value, color = Index)) +
```

```
  geom_line() +
```

```
  labs(
```

```
title = "Re-indexed European Stock Market Indices",
x = "Time",
y = "Index Value (First = 100)"
) +
theme_minimal()

# PART B: Dose-Response Curve

set.seed(123)
concentration <- c(0, 1, 2, 4, 8, 16, 32, 64)
root_length <- c(100, 96, 90, 82, 70, 55, 35, 18)

ryegrass <- data.frame(
  concentration,
  root_length
)
ggplot(ryegrass, aes(concentration, root_length)) +
  geom_point(size = 3) +
  geom_smooth(
    method = "loess",
    se = FALSE
  ) +
  labs(
    title = "Ryegrass Dose-Response Curve",
    x = "Concentration",
```

```

y = "Root Length"
) +
theme_minimal()
control <- ryegrass$root_length[ryegrass$concentration == 0]
half_control <- control / 2
difference <- abs(ryegrass$root_length - half_control)
result <- ryegrass[which.min(difference), ]
cat("Control root length:", control, "\n")

cat("50% of control:", half_control, "\n")

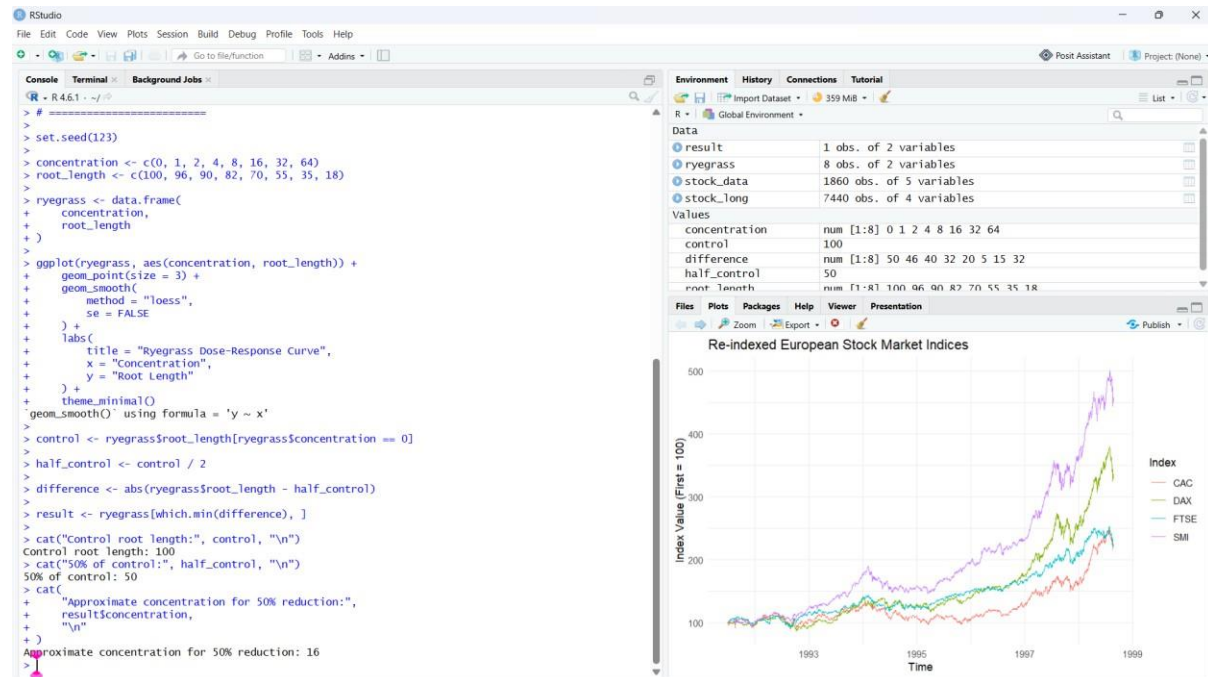
cat(
  "Approximate concentration for 50% reduction:",
  result$concentration,
  "\n"
)

```

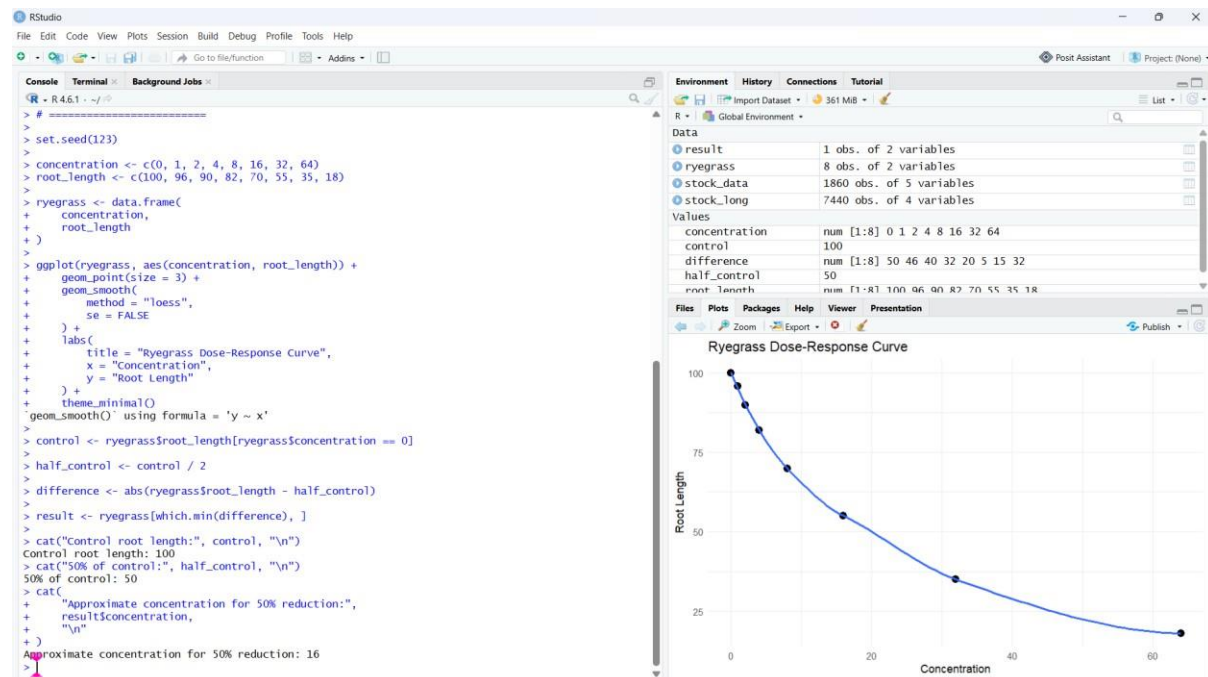
Interpretation

Output

Part a



Part b



Interpretation

Part A

- All indices are re-indexed to **100** at the starting point.
- This makes their relative growth directly comparable.
- The index ending at the highest value shows the strongest overall growth.
- The four markets show similar general movements but different growth levels.

Part B

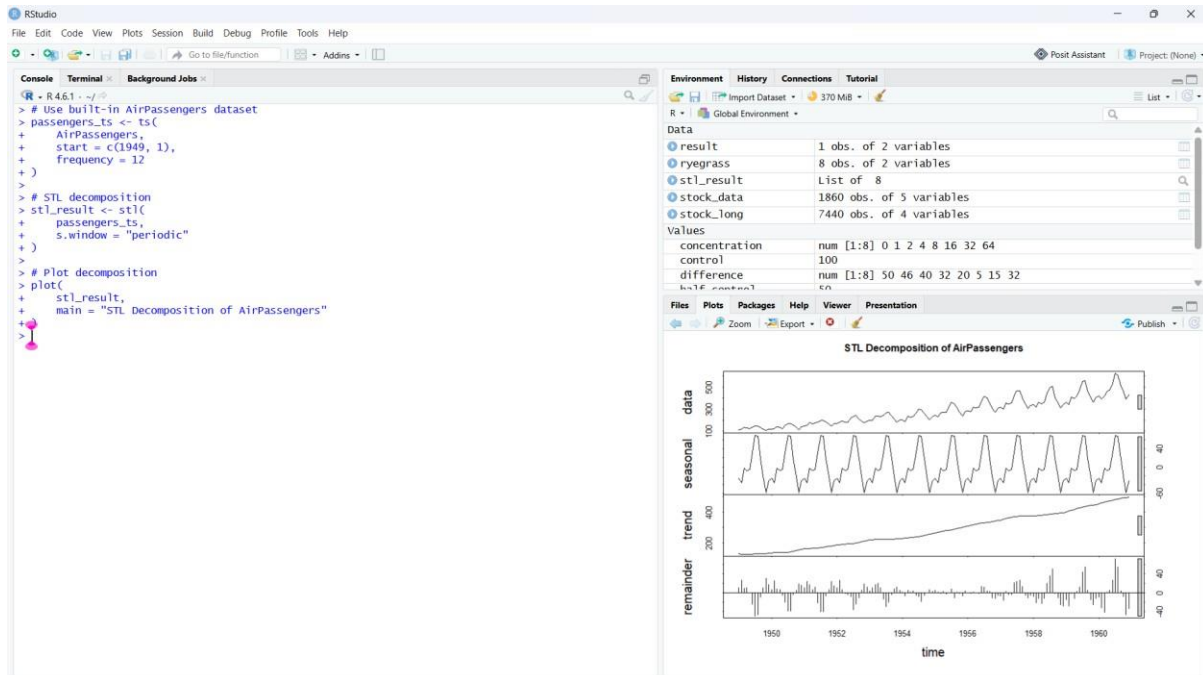
- Root length **decreases as concentration increases**.
- The curve shows a clear dose–response relationship.
- The control root length is **100**, so 50% reduction corresponds to **50**

QUESTION 3

Code:

```
# Use built-in AirPassengers dataset
passengers_ts <- ts(
  AirPassengers,
  start = c(1949, 1),
  frequency = 12
)
# STL decomposition
stl_result <- stl(
  passengers_ts,
  s.window = "periodic"
)
# Plot decomposition
plot(
  stl_result,
  main = "STL Decomposition of AirPassengers"
).
```


Output:



Interpretation

- The **trend component** shows a clear long-term increase in the number of passengers.
- The **seasonal component** shows a repeating yearly pattern, with passenger numbers increasing during certain months.
- The **remainder** represents irregular fluctuations that are not explained by trend or seasonality.
- The seasonal fluctuations become **larger as the overall passenger level increases**.

Therefore, a **multiplicative decomposition** is more appropriate because the seasonal amplitude grows with the trend

QUESTION 4

Code:

```
# CO2 time series
```

```
co2_ts <- co2
```

```
# Create time variable
```

```
t <- time(co2_ts)
```

```
# Fit linear trend
```

```
model <- lm(as.numeric(co2_ts) ~ t)
```

```
# Fitted trend
```

```
trend <- fitted(model)
```

```
# Detrended residuals
```

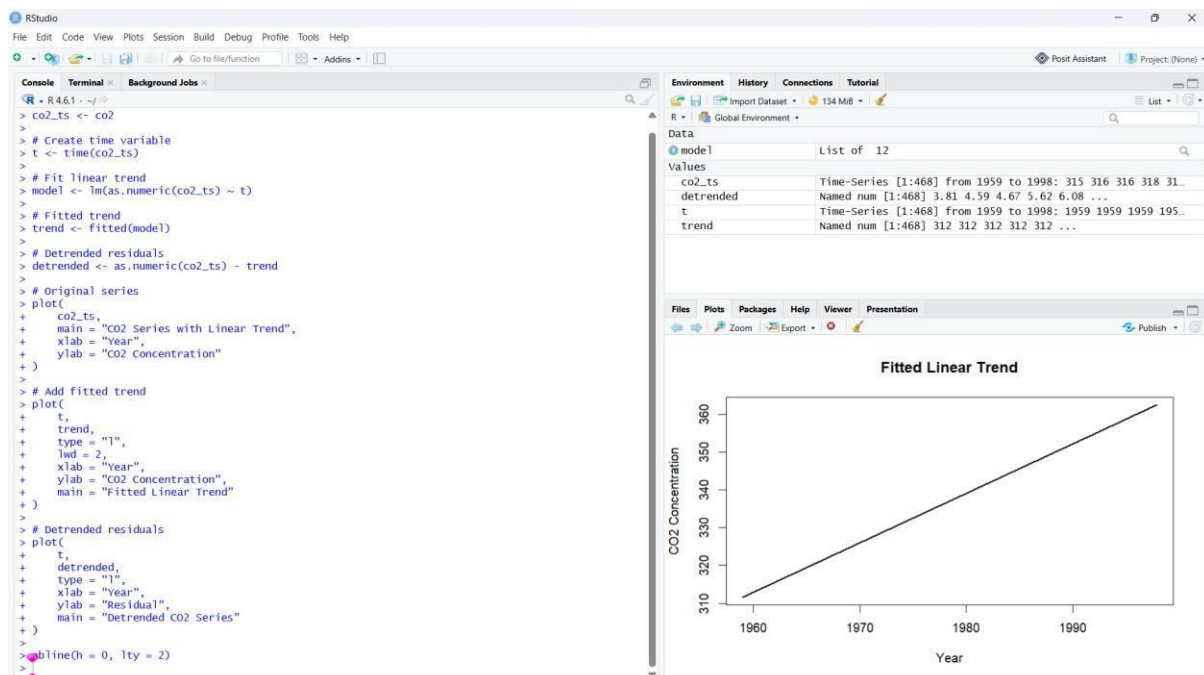
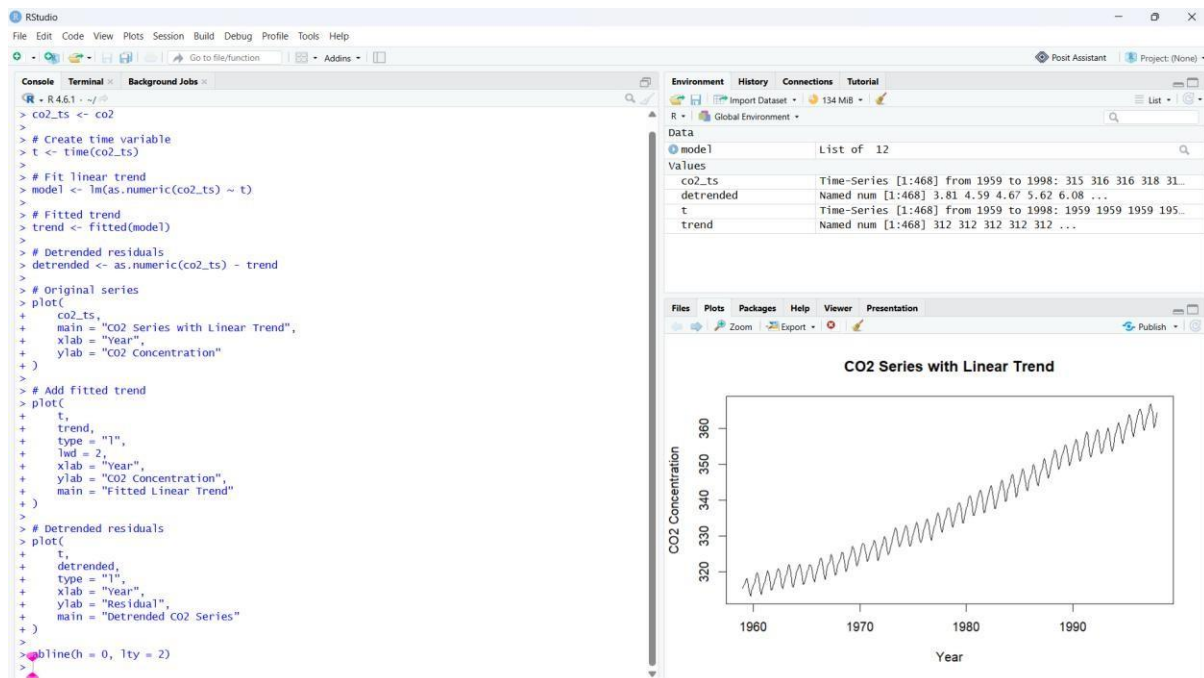
```
detrended <- as.numeric(co2_ts) - trend
# Original series

plot(
  co2_ts,
  main = "CO2 Series with Linear Trend",
  xlab = "Year",
  ylab = "CO2 Concentration"
)
# Add fitted trend

plot(
  t,
  trend,
  type = "l",
  lwd = 2,
  xlab = "Year",
  ylab = "CO2 Concentration",
  main = "Fitted Linear Trend"
)
# Detrended residuals

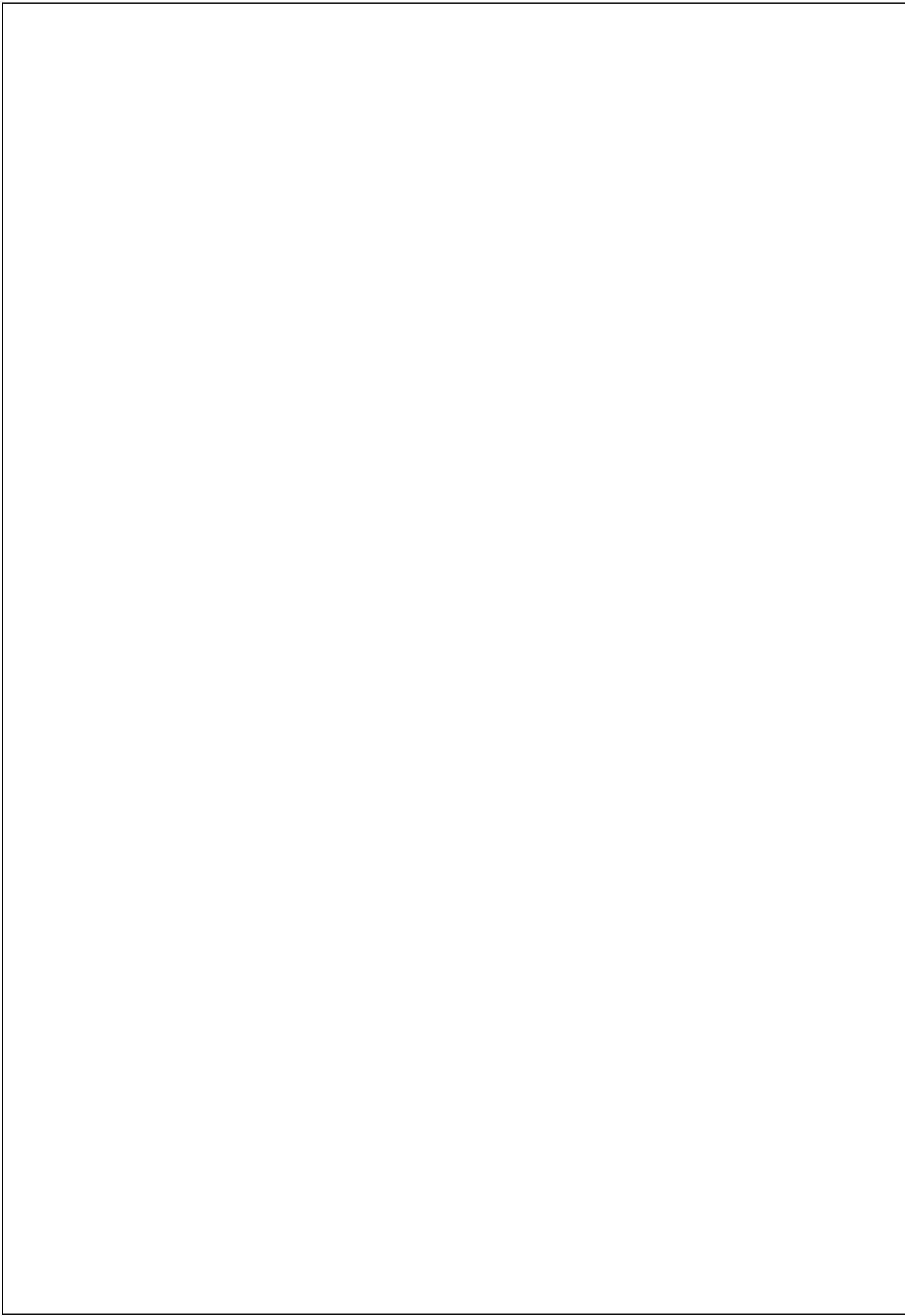
plot(
  t,
  detrended,
  type = "l",
  xlab = "Year",
  ylab = "Residual",
  main = "Detrended CO2 Series"
)
abline(h = 0, lty = 2)
```

Output:



Interpretation

- CO₂ shows a strong **long-term increasing trend**.
- The detrended series still contains a clear **repeating seasonal pattern**.
- CO₂ rises and falls regularly within each year.
- Linear detrending removes the overall trend but **does not remove seasonality**.



QUESTION 5

Code:

```
# Install once if needed
install.packages("forecast")

library(forecast)

# PART A: Forecasting Trends

# AirPassengers time series
air_ts <- AirPassengers

# Fit ETS model
model <- ets(air_ts)

# 24-month forecast
forecast_result <- forecast(model, h = 24)

# Plot forecast
plot(
  forecast_result,
  main = "24-Month Forecast of AirPassengers",
  xlab = "Year",
  ylab = "Number of Passengers"
)

# =====
```

```
# PART B: Cycle Detection
```

```
# =====
```

```
# Sunspots annual data
```

```
sunspot_ts <- sunspot.year
```

```
# Plot ACF
```

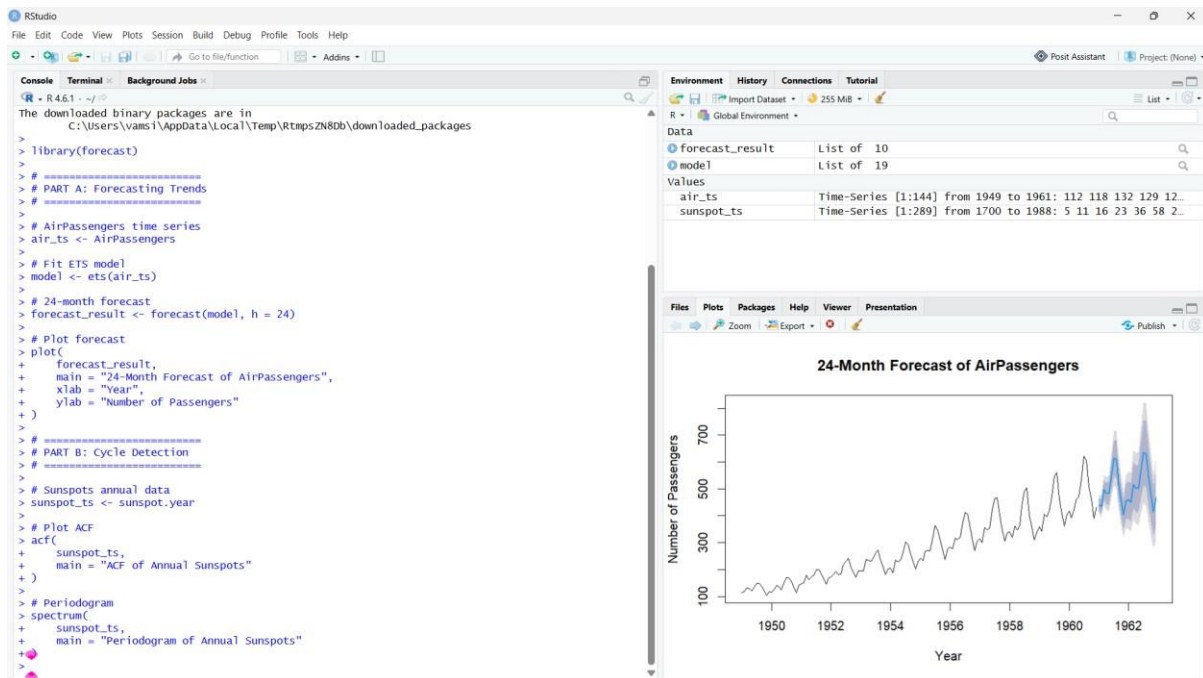
```
acf(  
  sunspot_ts,  
  main = "ACF of Annual Sunspots"  
)
```

```
# Periodogram
```

```
spectrum(  
  sunspot_ts,  
  main = "Periodogram of Annual Sunspots"  
)
```

Output:

Part a — Forecasting Trends



Part b — Cycle and Seasonality Detection

